Parametric Tools for structural BIM Design

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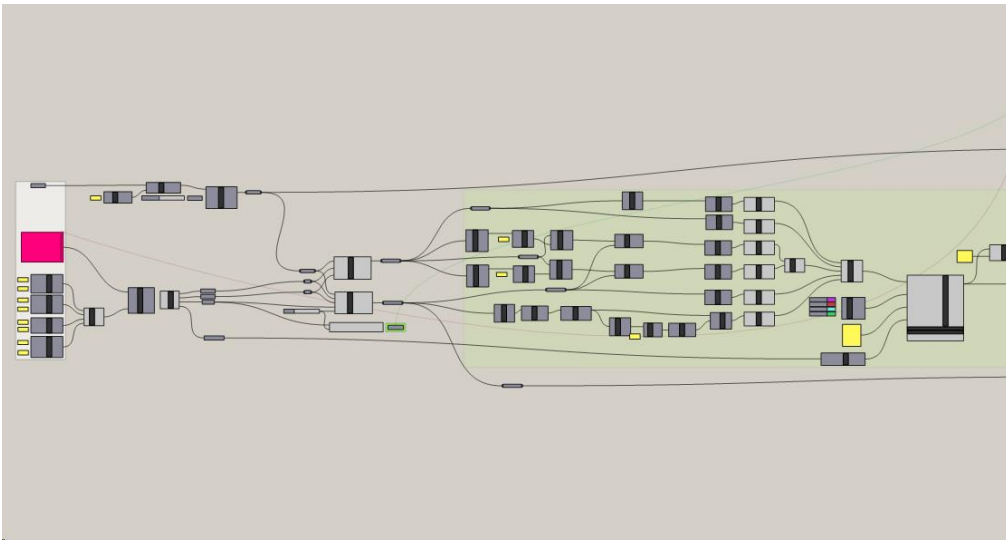


Fig. 1: Screenshot of final Grasshopper Script

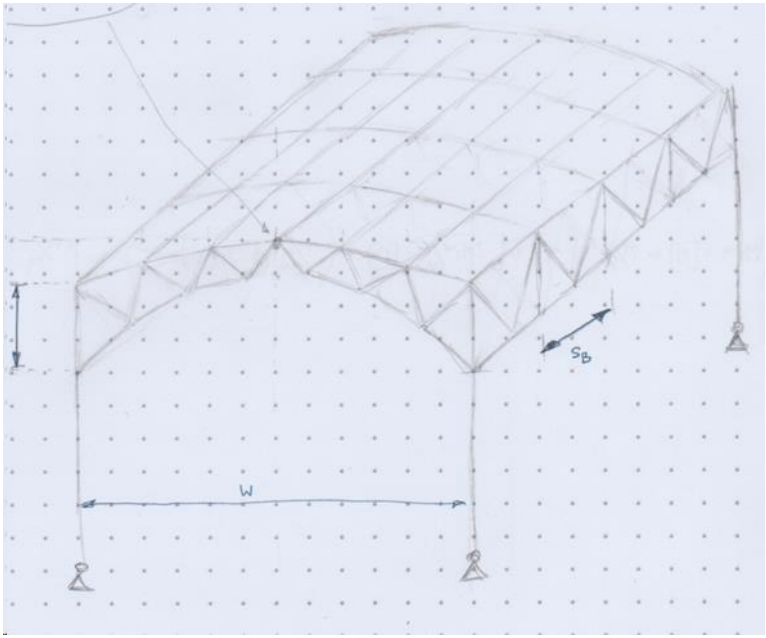
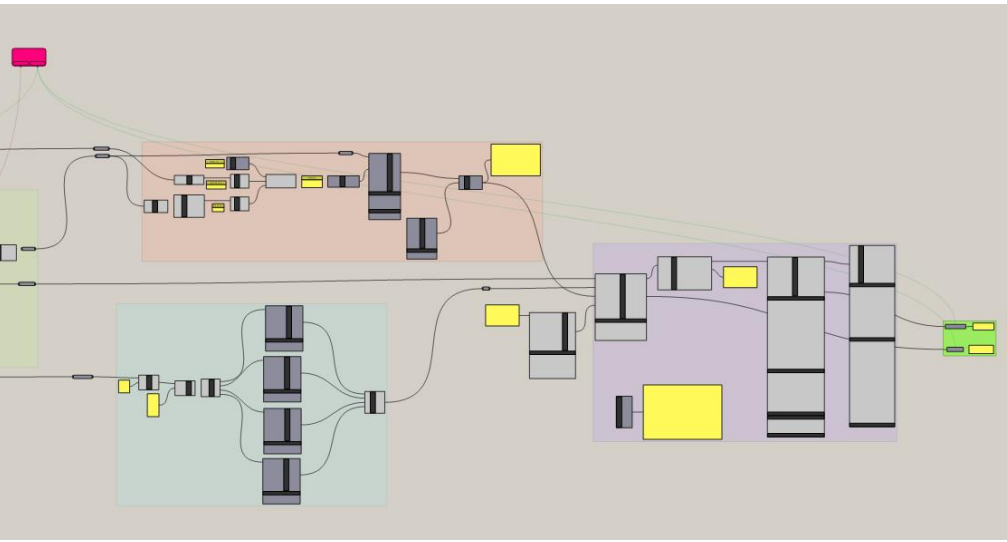


Fig. 2: Initial design sketch



Brief discription / Abstract

The objective of this project is to design a roof structure for a public space. The coverable area has a dimension of 10x10m spanning a total of 100m² of which at least 40% need to be covered. Vertical limits contraint the design within a bounding range between 4–8m above ground level.

The Structure is to be modeled as a volumetric truss and its shape can be freely chosen by the designers. The main construction material for the structural truss is timber. Materials of the skin and columns are not specified.

All modeling has to happen entirely using Rhino and grasshopper, with appropriate parametrization of key properties. Karamba3D is used for structural analysis/FEM, and the Octopus plugin is used for multi-objective optimization of the chosen parameters.

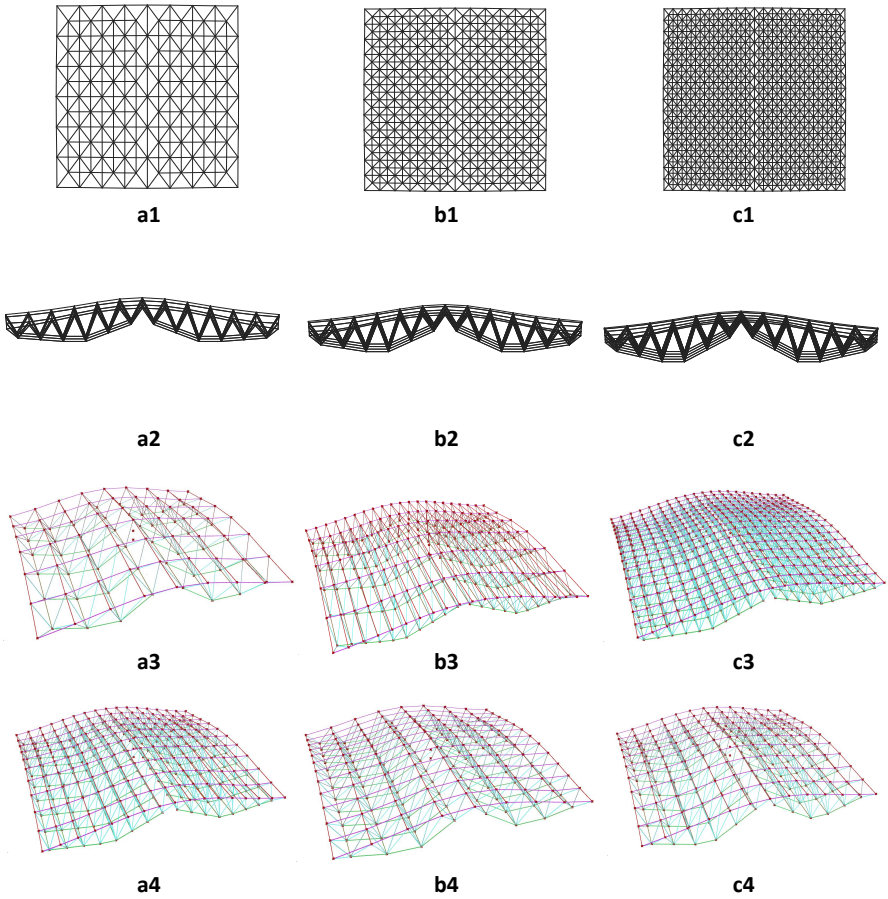


Fig. 3 : Results of parameter variation

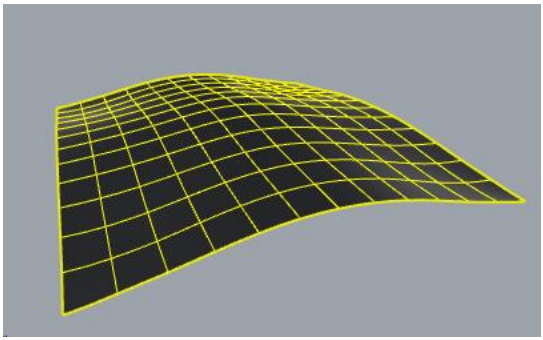


Fig. 4 : Reference Surface

Parametric structure (concept)

The entire structural component of the roof is designed as a spatial truss made up of timber beams. Along the main-axis the static system reflects a three-hinged frame with a central hinge at the symmetrical axis of the roof and two at the connection points between roof and the columns. The columns are assumed as vertical cantilever beams, but not explicitly part of the structural analysis. Hence their material properties and stiffness are not considered during analysis.

The static system of the truss was designed as an ideal truss, allowing for full rotational freedom at all hinges along the local Y and Z beam axis.

Shape:

The shape of the roof's skin was modeled manually within Rhino and then used as the basis for parametric modelling of the supporting truss structure. Besides aesthetics, a main consideration of the final shape is concentration of water run-off paths towards the columns. Rain- and meltwater is naturally collected towards the corner points of the structure without the need of any gutters. This eliminates need for regular maintenance and allows vertical water-discharge alongside the columns. Ideally this water run-off can be used to water planters/green islands surrounding each column, improving seepage and on-site rainwater use.

Materials:

The truss beams were modeled with timber beams of type "Duo/Trio Balken". These construction-grade beams offer great durability, consistent straightness and acceptable cost. As part of the parametric optionality, the catalog from commercial supplier "J.u.A Frischeis GmbH" was manually encoded into the Grasshopper script. The hinges will be implemented using industry standard steel connections and are not explicitly detailed as part of this analysis and report.

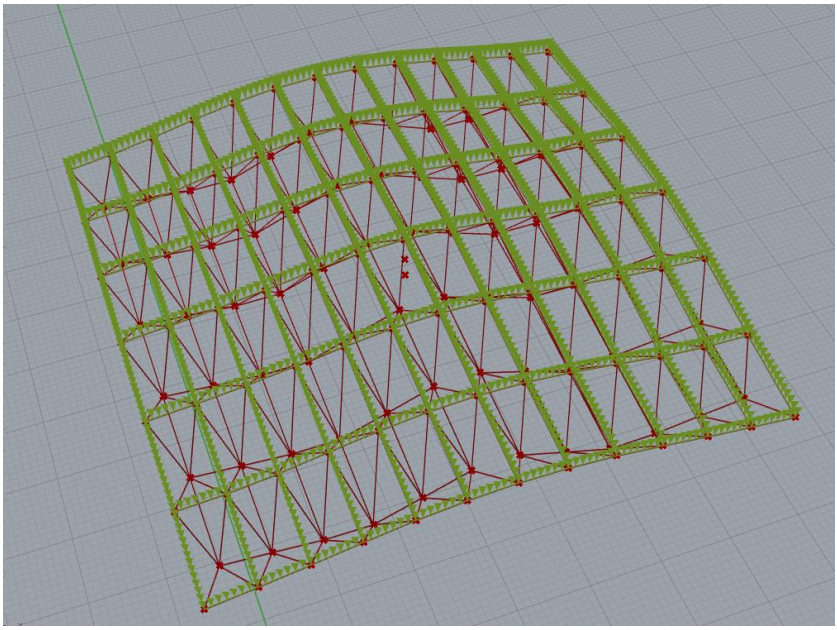


Fig. 5: Vertical Load (green arrows)

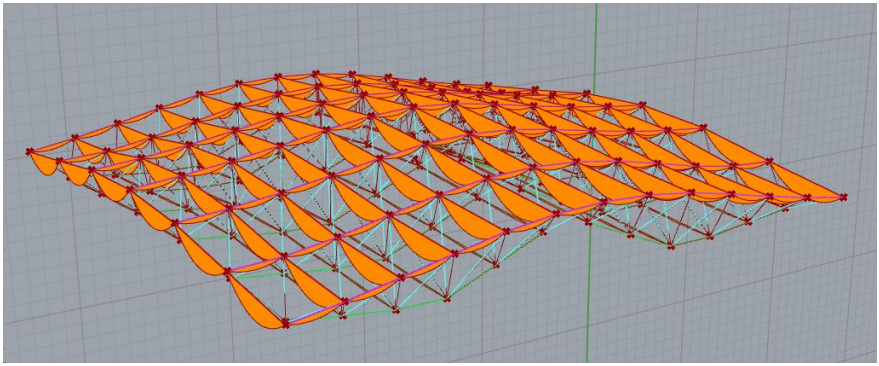


Fig. 6 : Reactions, My

Criterias and Objectives

Overall the entire structure ought to appear lightweight. This is achieved by using smaller cross sections for the beams and empty space between beams/hinges.
These goals compete with structural goals, e.g. minimal deflection and reduced buckling. The following objectives capture these considerations:

- **Min. total mass**
- **Max. ratio of highest to lowest truss height**
- **Min. deflection of beams**

<u>Parameters:</u>
Variability for the optimization process was given by parametrizing:

- **Number of truss segments along the global X-axis**
- **Number of truss segments along the global Y-Axis**
- **Beam cross section**
- **Height-Ratio**

The Cross section could also have been optimized using a *dedicated Karamba component Cross Section Optimizer,* but as part of this project was manually optimized by providing the cross-section selection as a Gene/Parameter to Octopus

<u>Global Width/Depth:</u>
The author chose a fixed overall dimension for the structure's global width and depth. This trivially satisfied the required covered area of 40%. It also enables other stakeholders, e.g. gardeners, a more concrete idea of the final cover area, providing more certainty and for better planning. The GH script contains a slider to change the overall dimension of the roof, if it needs to be adjusted or later added as a parameter for optimization.

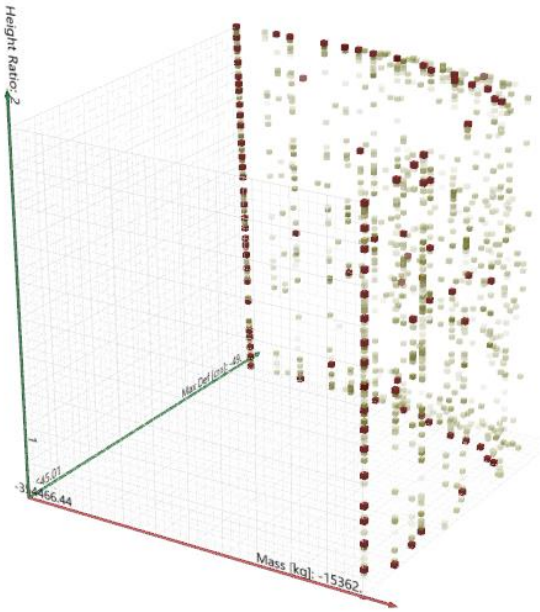


Fig. 7: Octopus visualisation of calculated solution space

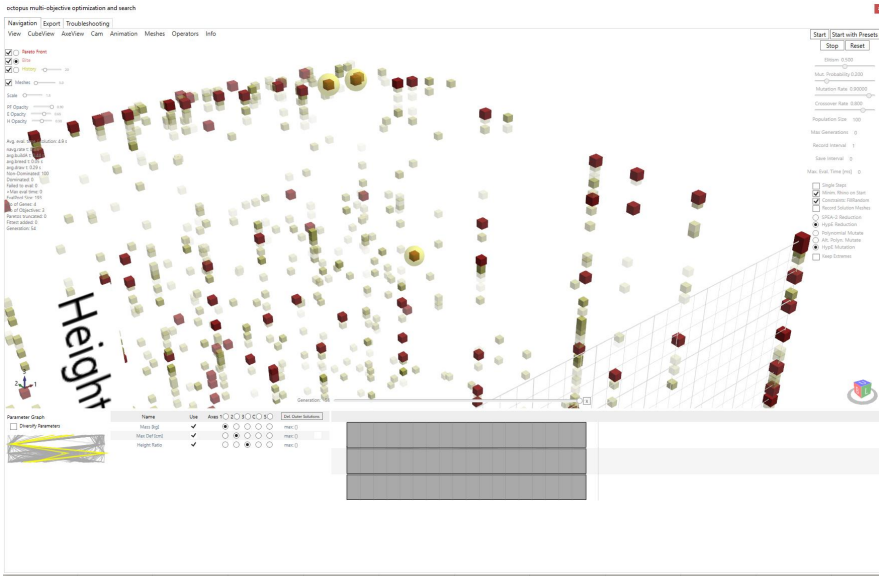


Fig. 8: Area of candidates for chosen candidate

Results and Discussion

The optimization with Octopus ran with a population-setting of 100 per generation. No limit on the number of generations was given and the calculation was manually stopped after about 3 hours. Fig. 7 shows the resulting solution space.

From the set of possible candidates, the following final parameters were chosen:

- **X-segments: 0.31 (12 segments)**
- **Y-Segments: 0.00 (6 segments)**
- **Height Ratio: 0.33 (2.44)**
- **Cross section: 0.19 (14/14 cm)**

This parameter-combination did *not* result in a total maximum of all given objectives: For pragmatic reasons, a combination with slightly less optimal values was chosen. Too many X and Y segments would be practically difficult to manufacture and assemble due to many differently sized beams and high number of joint connections.

As possible improvement, such consideration can be encoded in a further iteration of this script. Such an approach exceeded the scope of the present project.

The initial design proposal, Fig 2, proved antithetical to the idea of parametrising the height(-ratio): Due to the static model of a three-hinge frame and the implicit hinge at the column connection, no moment would be present at the outer edges of the idealised truss-beam. Hence the initially high truss edge did not add any advantage to the load bearing properties of the truss and was always "optimized away" by octopus, i.e. optimal solutions would always have a low height. This resulted in a change of design design with only high trusses towards the center of the idealised truss-beam between two hinges, where the moment is most significant.